SIMATS ENGINEERING

ASSESSMENT TOOL 1: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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Register Number	**192525076**

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I **R.Pranathi(192525076)** (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pranathi

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

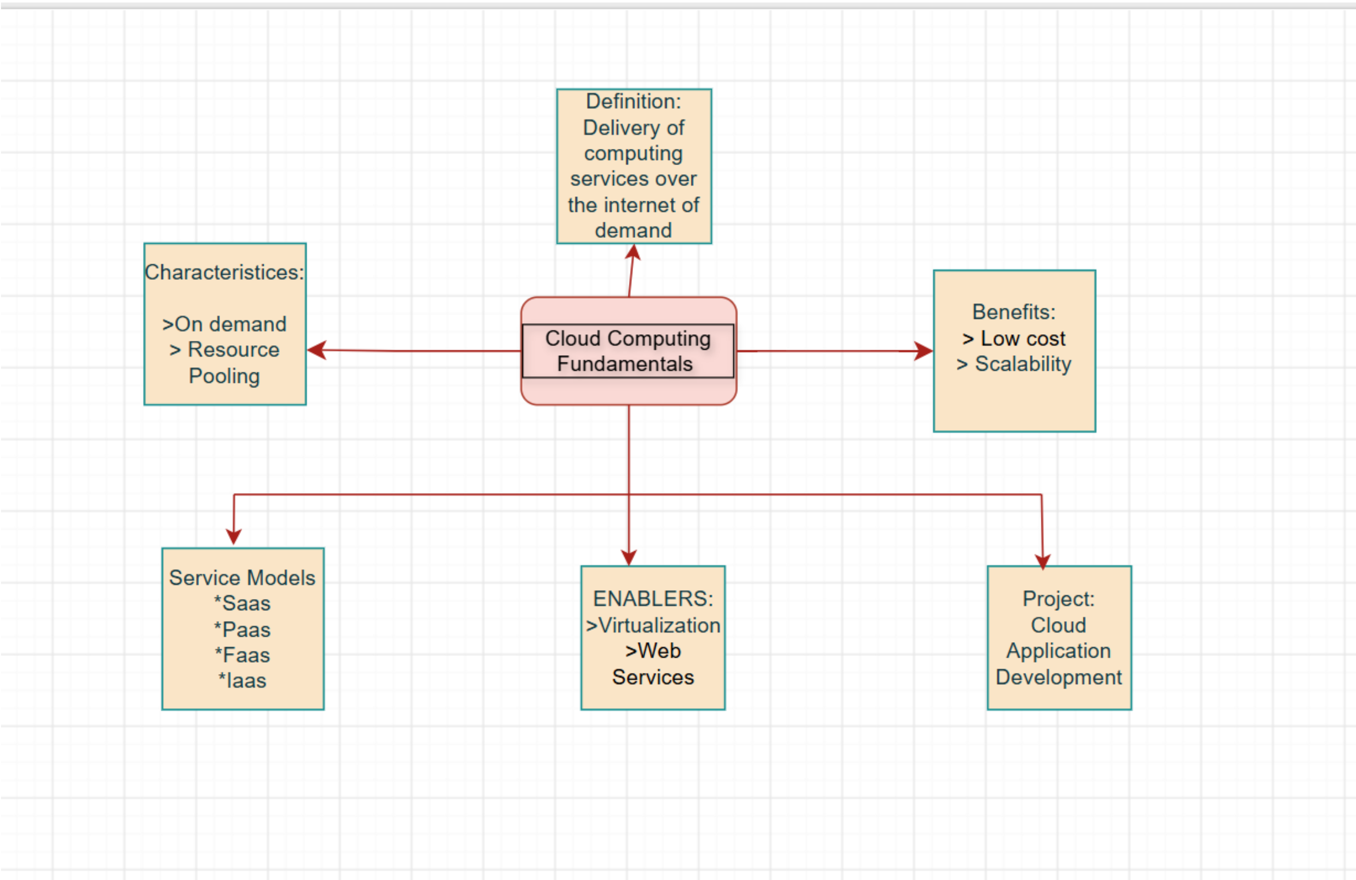
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

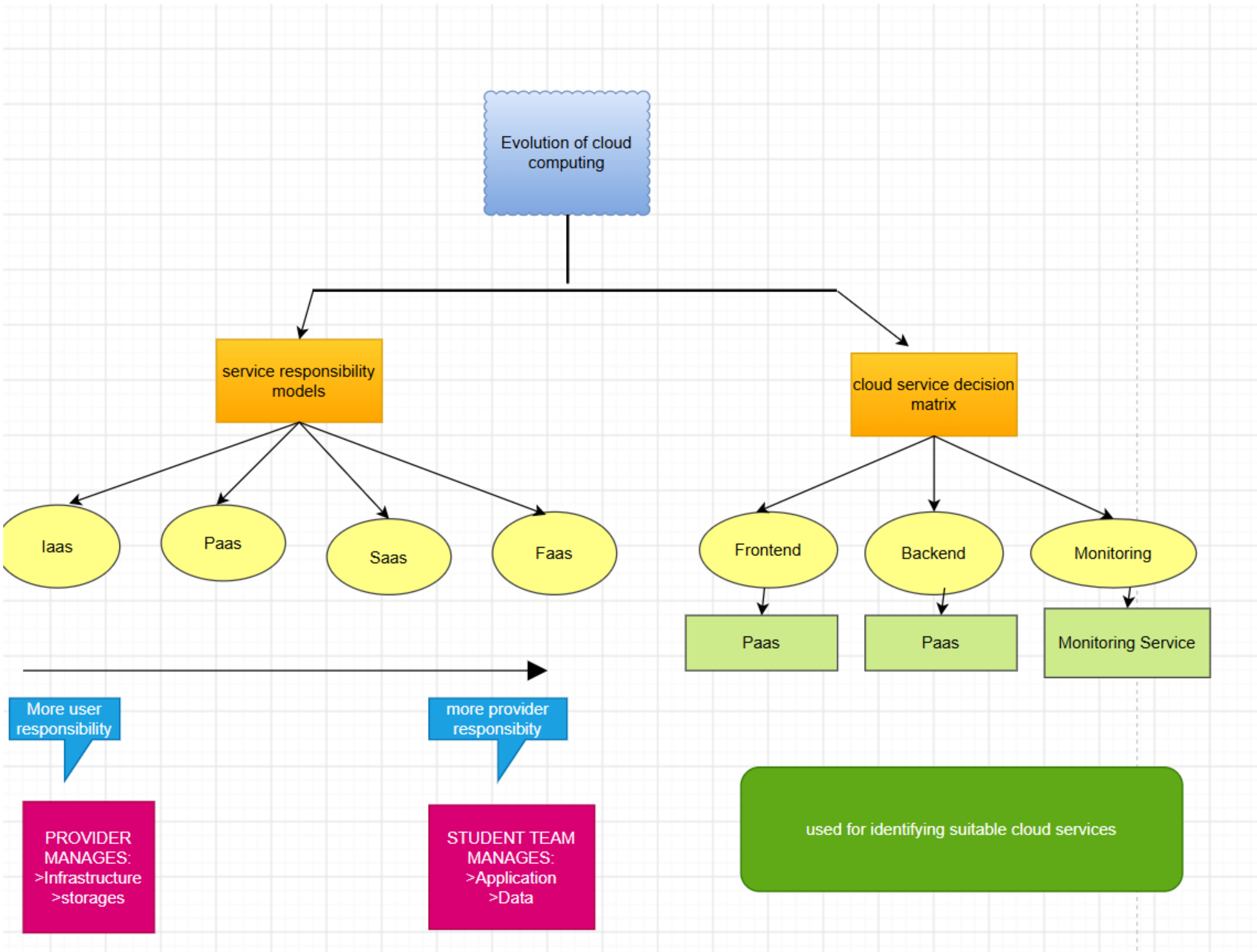
5. Questions

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?
ii). Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?
iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?
iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?
v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

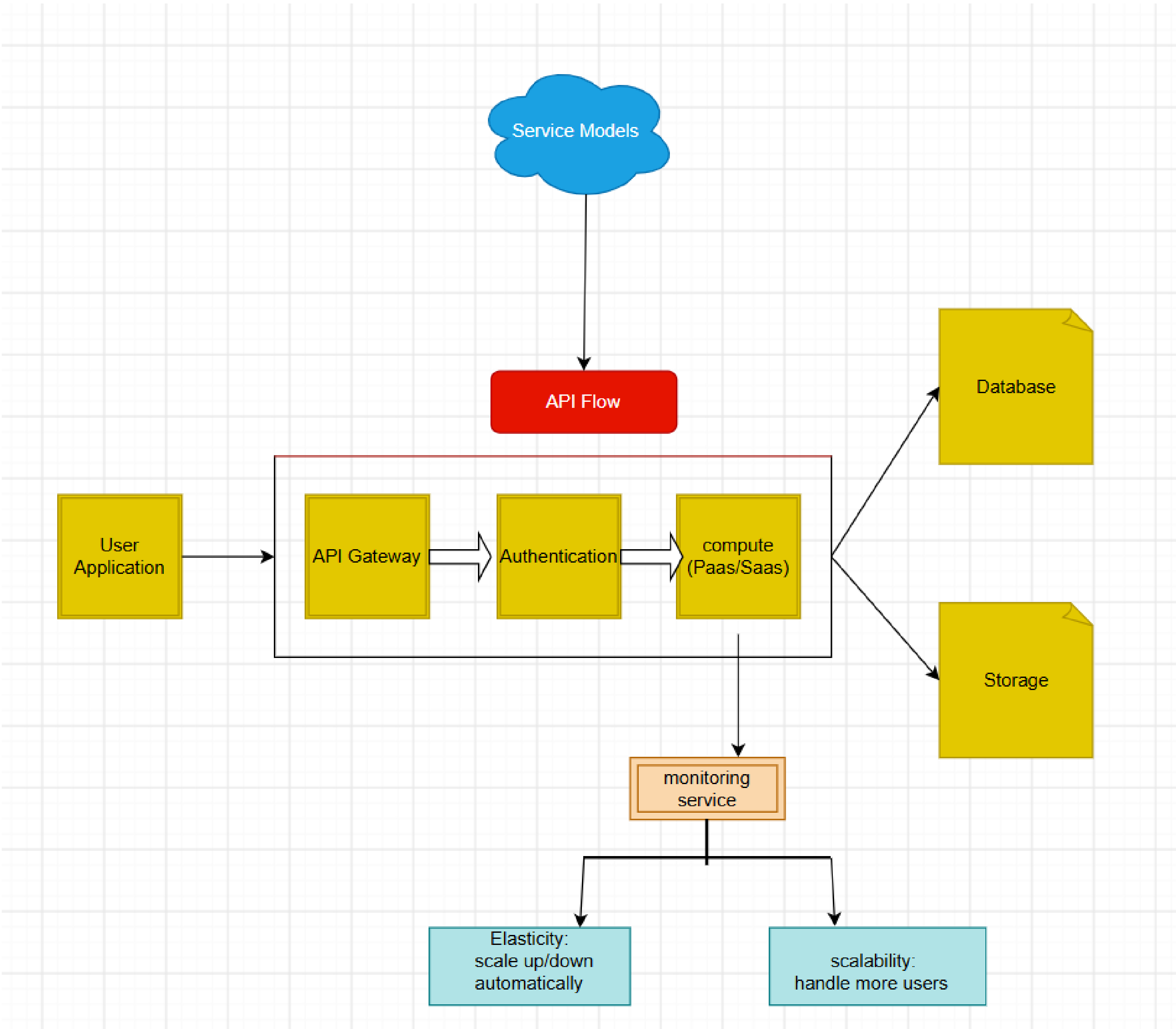
Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.



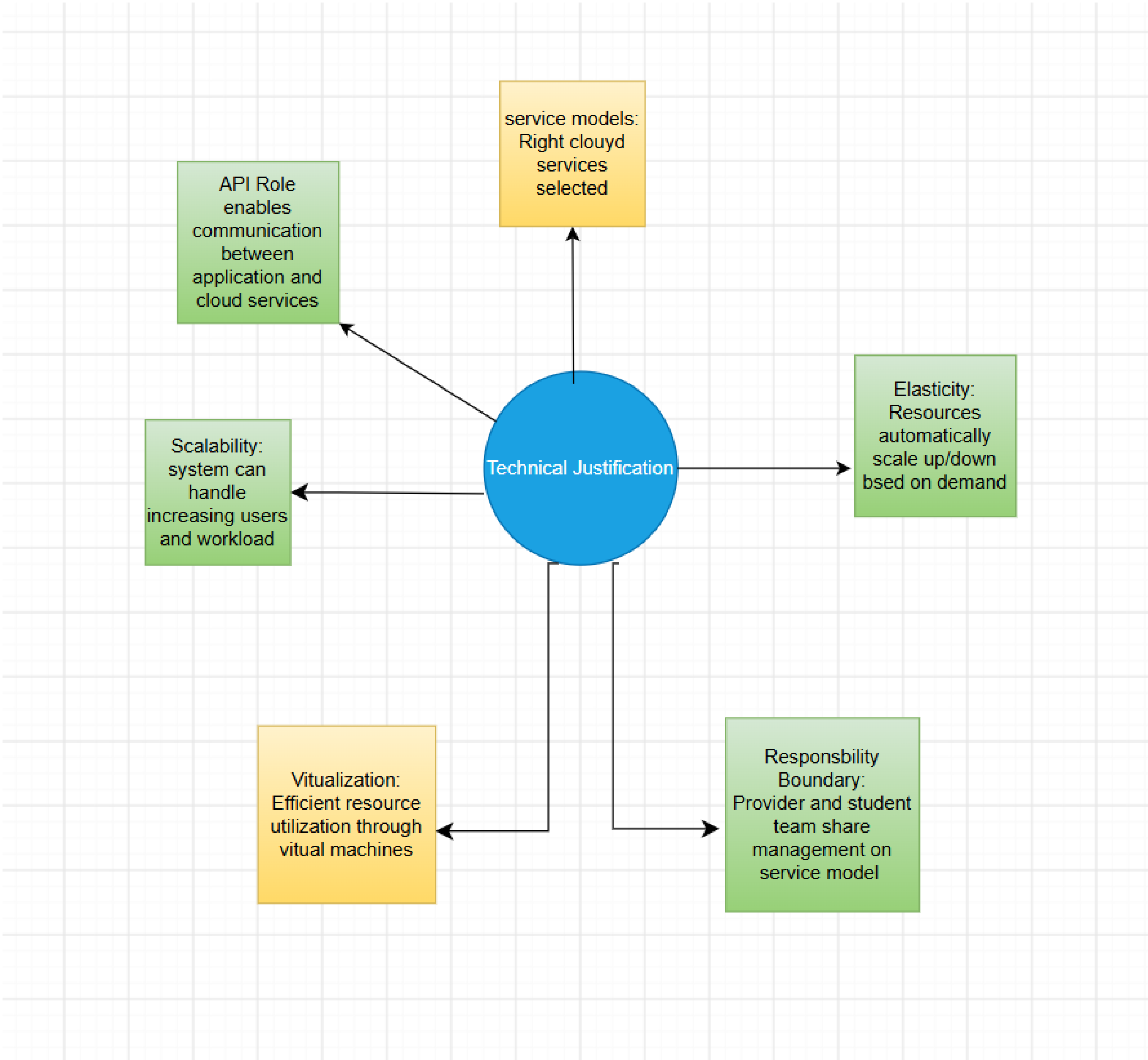
Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.



Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.



Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.



QUESTIONS AND ANSWERS

i). Evaluate the effectiveness of your concept map in representing cloud computing principles. Which design decisions most strongly support your solution, and why?

ANS:

- It represents cloud computing by showing the relationships between users, cloud services, deployment models, and infrastructure.
- The use of clear connections and hierarchy makes the architecture easy to understand.
- Grouping related components such as SaaS, PaaS, and IaaS helps explain service models clearly.
- These design decisions improve readability and demonstrate how cloud components work together.

ii) Analyze how scalability, elasticity, and resource management influence the architecture shown in your concept map. What challenges could arise if these capabilities were absent?

ANS:

- Scalability allows the system to handle increasing workloads by adding resources.
- Elasticity automatically adjusts resources based on demand, reducing cost and improving efficiency.
- Resource management ensures optimal allocation of computing resources.
- Without these features, the system could experience slow performance, resource wastage, downtime during high traffic, and increased operational costs.

iii). Justify the selection of the APIs, web services, or cloud services included in your concept map. How do they contribute to the overall functionality and efficiency of the system?

ANS:

- APIs enable communication between applications, while web services allow data exchange across platforms.
- Cloud services such as SaaS, PaaS, and IaaS provide software, development platforms, and infrastructure on demand.
- Together, they improve system integration, reduce development time, increase flexibility, and ensure efficient resource utilization.

iv). Critically examine the relationships between the components in your concept map. How do these interactions improve reliability, performance, or user experience?

ANS:

- The components interact through APIs, virtualization, storage, networking, and cloud services to provide seamless communication.
- Virtualization improves resource utilization, while cloud storage ensures data availability.
- Load balancing and scalable infrastructure improve performance and reliability.
- These interactions provide users with fast, secure, and uninterrupted access to cloud applications.

v). Reflect on the process of developing your concept map. How has creating and analyzing the map changed your understanding of cloud computing architecture, and what improvements would you make if you were redesigning it?

ANS:

- Developing the concept map improved my understanding of how cloud computing components are interconnected.
- It helped me visualize the relationships between service models, deployment models, and cloud technologies.
- If redesigned, I would include more security features, real-world cloud providers, and additional details on networking and data flow to make the concept map more comprehensive and practical.